

Summary

Data Scientist with an **MS in Data Science** and current **MS in Computer Science** (online) candidate. Proven track record in developing **end-to-end ML solutions**, including **NLP**, **Computer Vision**, and **Generative AI**. Expert in **Python** and **SQL**, with deep experience in **MLOps (CI/CD, Docker)**.

Education

- **Georgia Institute of Technology** 2025 Sep - Present
 - Master's Degree in Computer Science (*Online*)
 - Relevant Courses: Machine Learning for Trading, Database System Concept and Design, Data and Visual Analytics
- **New York University** 2021 Sep - 2023 May
 - Master's Degree in Data Science
 - Relevant Courses: Machine Learning, Applied ML in Finance, NLP and Computational Semantics, Deep Learning, Time Series Analysis, Probability & Statistics, Big Data
- **Shandong University** 2014 Sep - 2021 May
 - Master's Degree in Economics/Bachelor's Degree in Management Science & Engineering
 - Relevant Courses: Advanced Micro & Macro Econ, Econometrics, Finance, Calculus, Linear Algebra, Data Principles and Applications, Python and Data Analysis, Database Management

Experience

- Data Scientist, Marketing @Tractor Supply Co.** Brentwood, TN | 2025 Jan - Present
- **Scalable Data Engineering:** Integrate 1M+ daily records (site traffic, financial) into **Snowflake** using advanced **SQL** and **Git**-based workflows, ensuring high-performance data pipelines for downstream tasks.
 - **Business Intelligence:** Utilize **Power BI**, **Snowflake SQL**, and **Git**-based workflows to ensure automated, scalable reporting that translates complex data into actionable insights.
 - **A/B Testing & Analytics:** Lead the design and statistical analysis of **A/B tests** to quantify campaign effectiveness and provide marketing recommendations to cross-functional stakeholders.
 - **Cross-Functional Collaboration:** Partner with Marketing, Finance, and Merchandise teams to define key success metrics and standardize reporting frameworks, translating complex business objectives into actionable data-driven insights that inform budget allocation and performance management.

- Data Scientist, Global Automation & Engineering @Assurant** Nashville, TN | 2023 Feb - Jan 2025
- **ML & MLOps:** Develop **end-to-end ML** solutions to enhance overall performance in manufacturing workflows. This involves translating complex business requirements into scalable and data-driven ML applications, developing high-performing ML models, building and optimizing **CI/CD** pipelines to streamline the **deployment** and **maintenance** of ML solutions in **production environments** using tools such as **Docker**, **Azure ML**, **Mlflow**, **scikit-learn**, and **Git**. At Assurant, I successfully implemented an algorithm that integrates **NLP** and **Computer Vision** to automate USB debug mode on Android phones, resulting in a \$300k annual reduction in labor costs and generating additional revenue through external sales.
 - **Data Analysis & Reporting:** Conduct data mining and data analysis to support strategic decision-making processes for **VP-level** leadership. This involves utilizing tools such as **Python**, **Pandas/Numpy**, **SQL**, **PySpark** to perform in-depth data analysis on large-scale datasets, developing **real-time data visualization apps** that not only streamline repetitive data analysis and cleaning tasks but also effectively translate complex insights into actionable recommendations through dynamic and intuitive graphical representations. At Assurant, I led the development of real-time data apps that integrate, analyze, and visualize warehouse data from various sources (e.g., **databases**, **APIs**, and **IoT** devices) using tools like **Streamlit**, **Django**, **JS**, **CSS**, and **HTML**. These apps now empower users, including warehouse operators and executives, to access critical insights quickly, facilitating informed decision-making across the organization.
 - **Collaboration & Communication:** Lead cross-functional brainstorming sessions to address challenges using data-driven solutions, fostering a culture of teamwork by encouraging open dialogue and sharing of ideas across departments. I also utilize project management tools (e.g., **JIRA**) to track progress, assign tasks, and maintain clear communication across teams, ensuring timely delivery of data initiatives that align with organizational goals.

- Research Assistant (Automatic Speech Recognition, ASR), Center for Data Science@NYU** New York, NY | 2022 Sep - 2023 May
- **Data Preprocessing:** Employed various data cleaning and augmentation techniques to transform raw audio data (375 hours) into high-quality inputs for speech recognition models for effective model training and evaluation.
 - **NLP & ASR:** Fine-tuned an ASR model on both labeled and pseudolabeled Holocaust testimony audio data using **PyTorch**, followed by the inclusion of a word-based Language Model in final search, drove down the Word Error Rate to 13.5%, a huge gain in performance. **Weights & Bias (W&B)** and **High-Performance Computing (HPC)** are utilized to track experiments and accelerate the training process respectively.
 - **Academic Writing:** Honed my skills in articulating complex ideas clearly and collaborating on research initiatives, as demonstrated by my publication at the 2023 *Interspeech conference*.

- Data Science Intern, Digital Transformation@Campana & Schott** New York, NY | 2022 May - 2022 Sep
- **ETL:** Implemented robust ETL (Extract, Transform, Load) processes to streamline data collection and preparation. This included extracting data from various sources (**APIs**, **DBs**), transforming it into a usable format by cleaning and aggregating it, and loading it into centralized databases for further analysis or modeling.
 - **AI/ML:** Utilized advanced AI/ML techniques to address the complex challenges faced by pharmaceutical clients. This involved extracting meaningful insights from **unstructured** patient comments using **NLP** techniques such as **Named Entity Recognition (NER)** and **sentiment analysis**, employing **Time Series** and **Recurrent Neural Network (RNN)** models, such as **Long Short-Term Memory (LSTM)** networks and **N-HiTS** to capture historical sales fluctuation patterns, facilitating better strategic planning and resource allocation.
 - **Story Telling:** Developed comprehensive reporting frameworks to visualize and communicate insights to stakeholders effectively. This involved using tools such as **Power BI** to create interactive dashboards and reports that highlighted KPIs and trends, enabling stakeholders to make informed decisions based on real-time data analysis and facilitating ongoing monitoring of business metrics.

- Data Science Intern, R&D@iFLYTEK** China | 2019 May - 2019 Sep
- **Data Preprocessing:** Executed noise removal, audio file normalization, and data consistency checks to enhance large audio datasets, leading to more effective modeling for **Automatic Speech Recognition (ASR)**. This involved streamlining preprocessing tasks by developing automated scripts using **Python** and boosting the model's robustness and generalizability across various audio inputs by applying data augmentation techniques.
 - **Cross-Functional Teamwork:** Collaborated closely with researchers and engineers to analyze data patterns, identify challenges, and implement robust preprocessing methodologies, resulting in improved accuracy and performance of speech recognition models.

Certifications

- Power BI Data Analyst Associate
- Snowflake SnowPro Core
- Microsoft Azure Data Scientist Associate

Skills & Tools

- **Programming Languages:** Python, SQL
- **Machine Learning:** Regression, Classification, & Clustering, Recommendation Systems (e.g., RNN-based, DNN-based, Popularity-based, Collaborative Filtering, etc.)
- **AI:** NLP (Text Classification, Sentiment Analysis, NER, Chatbot, RAG), ASR
- **Visualization:** Power BI, Plotly, Dash
- **DBs:** PostgreSQL, MongoDB, MySQL, Vector DB, Azure Cosmos DB
- **Other:** Time Series Analysis, Langchain, Linux, Multithreading, Multiprocessing, Office, HTML, CSS, Git, FastAPI, PyTorch, Tensorflow, Mlflow, Flask, Django, Azure ML, Streamlit, Docker, Data Analytics, RESTful API, CI/CD, Version Control, ETL, JIRA, Snowflake, Adobe Analytics
- **Soft skills:** Stakeholder Engagement, Interdisciplinary Communication, Collaborative Problem Solving, Project Management, Feedback & Iteration, Cross-team Collaboration, Advanced Analytics, Supply Chain, Marketing Analytics